Illumina RNA and DNA library prep for mNGS



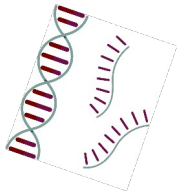
mNGS pipeline

Study design



- Type(s)
- Controls
- Storage
- Lab setup

xNA extraction



- Type(s)
- Controls
- QC

Library prep



- Host removal
- Fragmentation
- dsDNA generation
- End-repair
- Adapter ligation
- Barcoding & PCR
- Clean-up
- Pooling
- QC

Sequencing



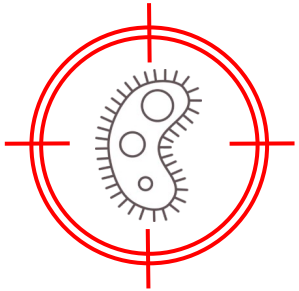
- Machine
- Read length
- Loading conc.

Data analysis



- QC data
- Host removal
- Alignment
- Blast
- Ranking of hits

Pathogen ID



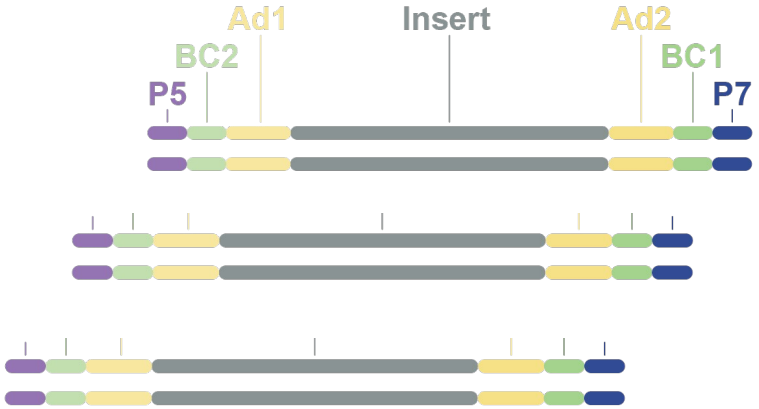
Library prep: RNA or DNA to sequencing library

- Sequencing library = Pool of DNA inserts with sequencing adapters attached



Extracted RNA or DNA

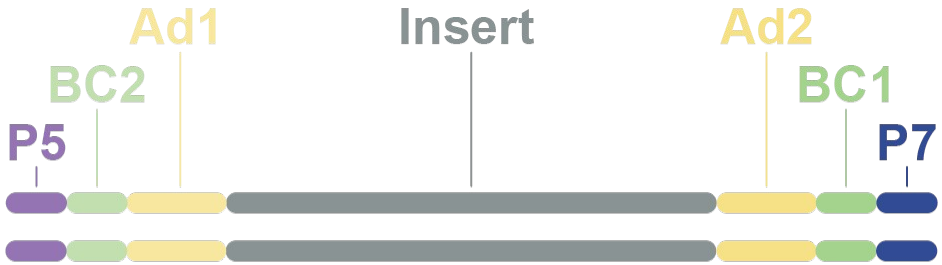




Sequencing library

What is a sequencing library?

- Sequencing library = Pool of DNA inserts with sequencing adapters attached



Insert = DNA fragment to be sequenced

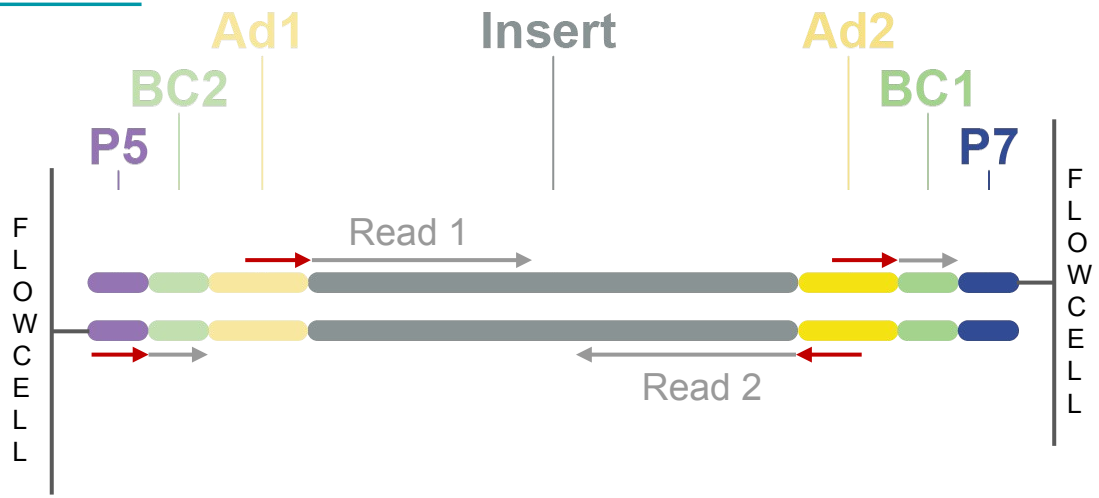
Ad1/Ad2 = Adapters that allow initiation of sequencing

BC1/BC2 = Barcodes specific for each sample (one for each strand)

P5/P7 = Flow cell binding sites

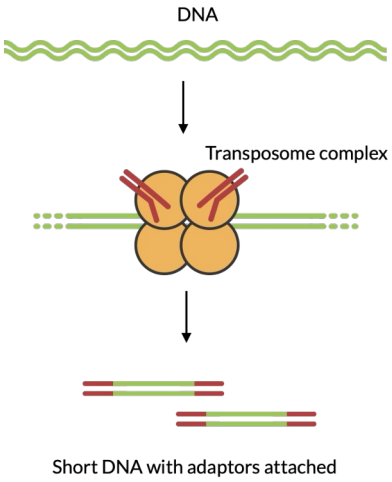
What is a sequencing library?

- Sequencing process:

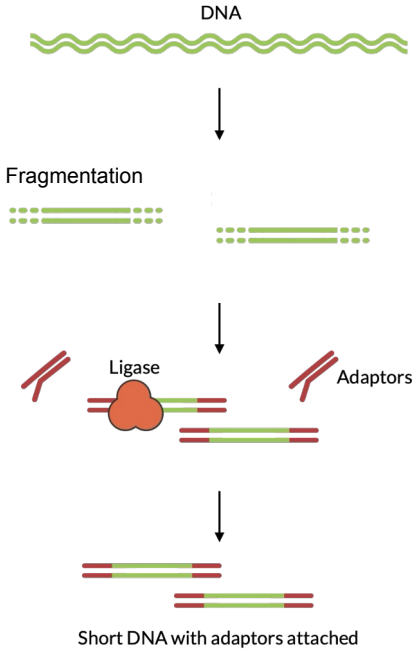


Two broad approaches to library preparation

Tagmentation
(fragmentation + adapter addition)

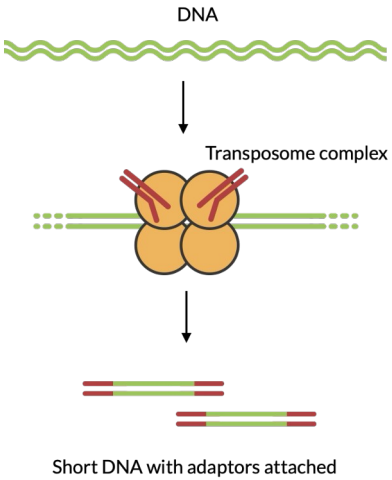


Fragmentation, then Adapter Ligation

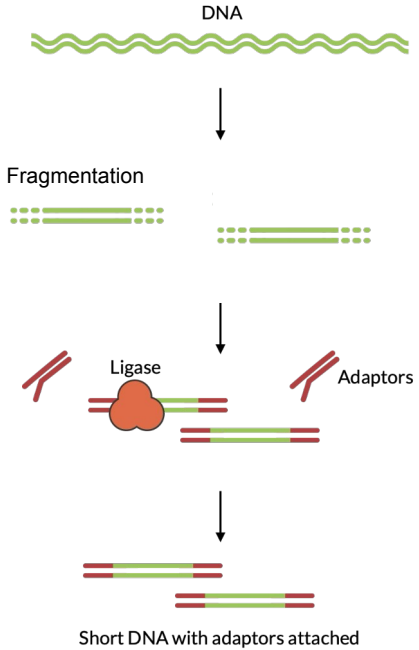


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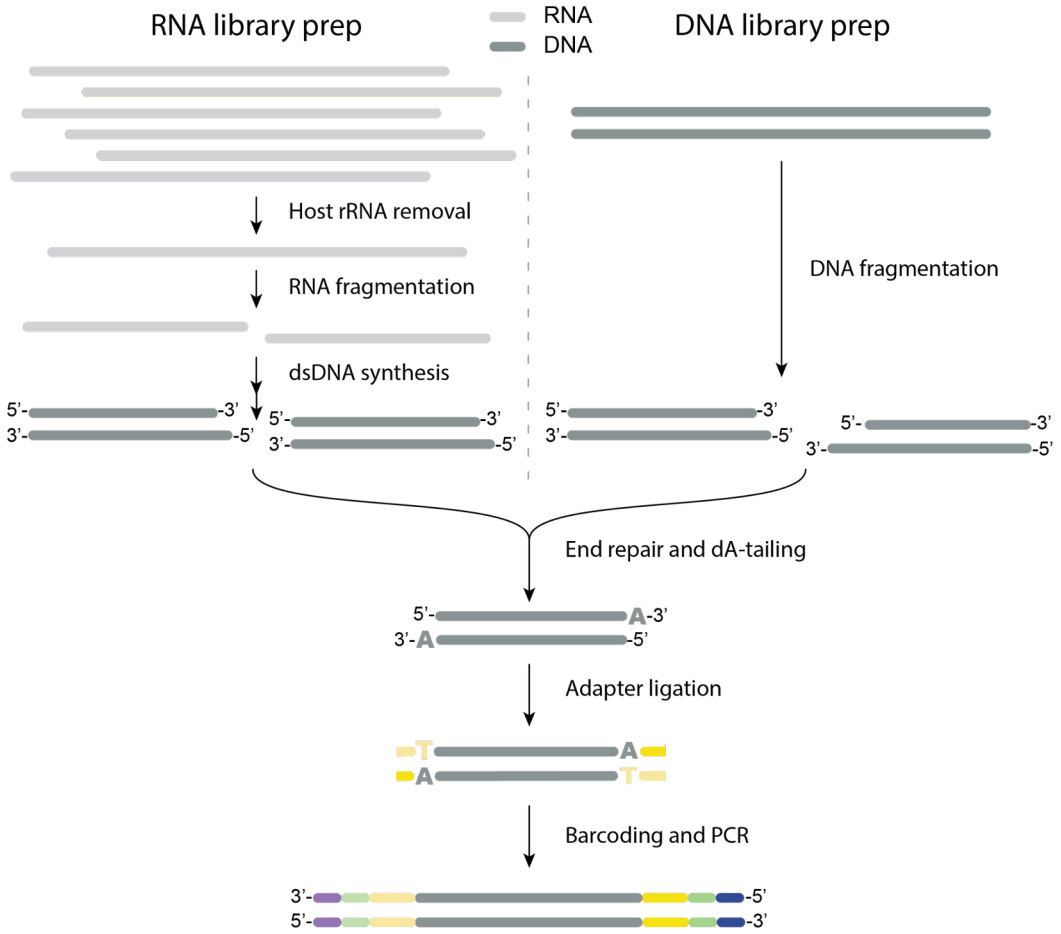


Fragmentation, then Adapter Ligation



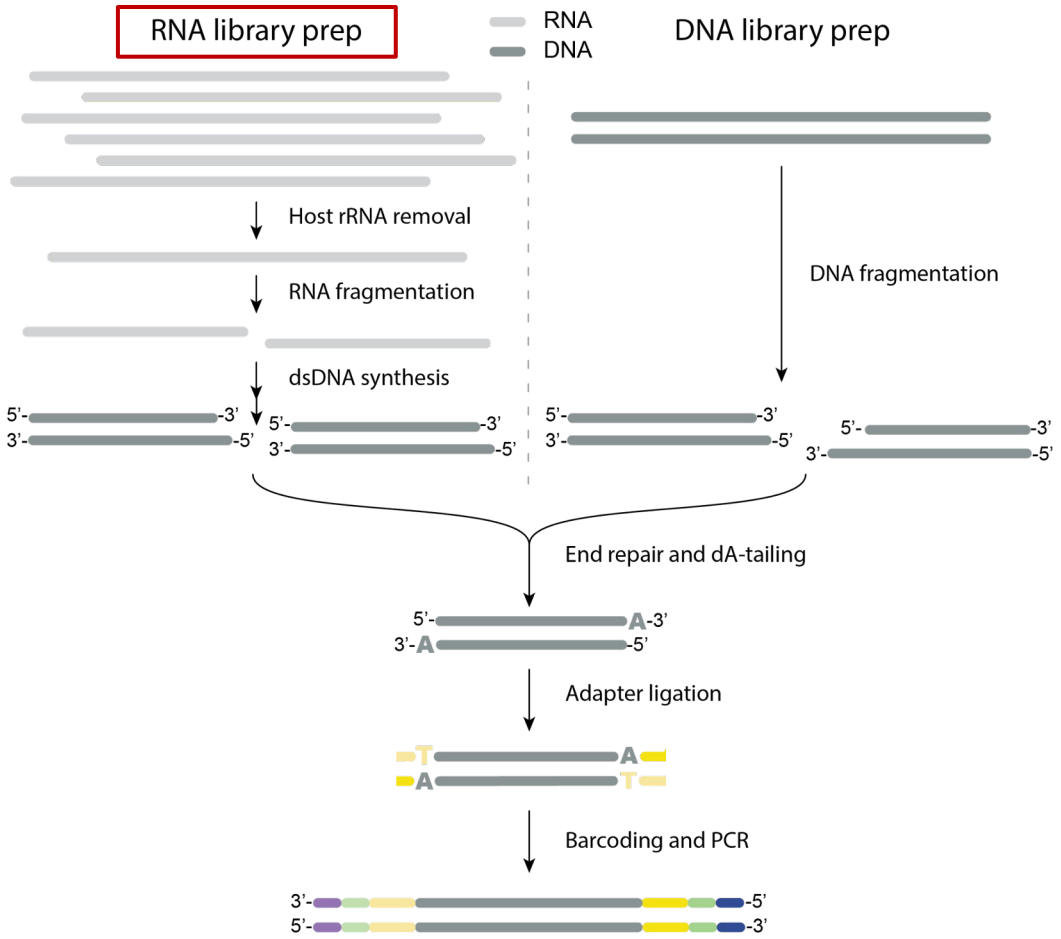
Library prep

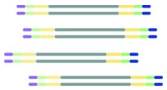
- Libraries are dsDNA



Library prep

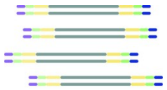
- Starting with RNA requires additional steps to convert it into dsDNA





Steps in library preparation:

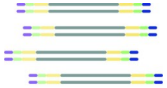
1. Host depletion, RNA fragmentation + priming
2. First strand cDNA synthesis
3. Second strand cDNA synthesis
4. End Prep
5. Adapter Ligation
6.. USER + Barcoding PCR



RNA library prep

STEP 0. Starting material: RNA or TNA

- RNA only: move on to host rRNA depletion
- TNA (total nucleic acid): need to remove DNA
  - Aliquot sample
  - DNAse I treatment
  - SPRI bead clean-up (2x)
  - Qubit quantification



RNA library prep

RNA content of typical host sample
95% host rRNA
2-3% host mRNA
1% other host RNA
<1% pathogen RNA

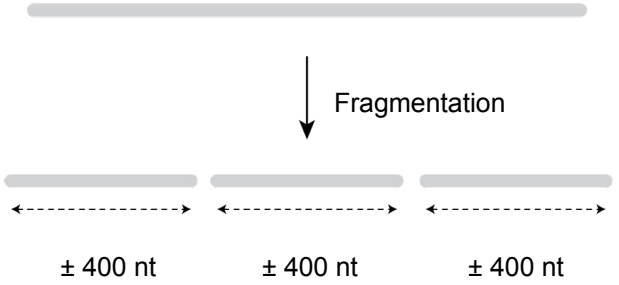
STEP 1. Host rRNA depletion

- Before RNA extraction:
  - Serum or plasma instead of blood
  - Supernatant of CSF to remove WBCs

- Human RNA depletion kits to use after RNA extraction:
  - FastSelect (Qiagen): $50 for 24 samples when reagent is diluted 1:10
    - Proprietary technology, presumably with hybridizing probes that make rRNA inaccessible to enzymes in subsequent steps

RNA library prep

STEP 2. Fragmentation



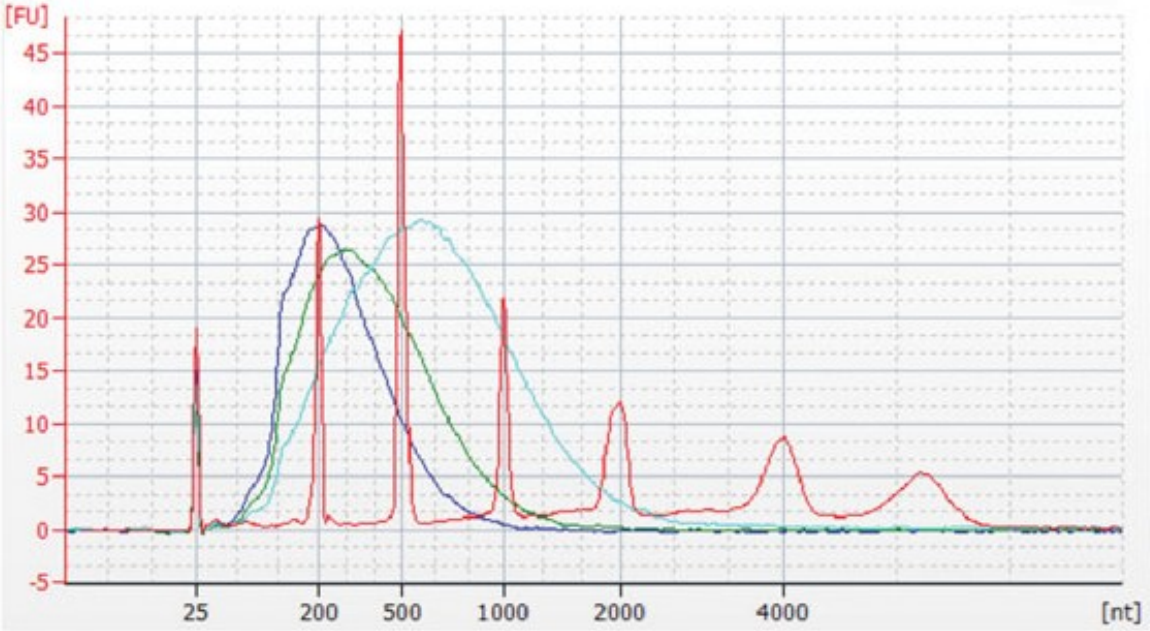
- Ideal insert size = ± 400 nt
- Several ways to fragment RNA:
  a. Enzymatic: %GC-dependent -> may bias mNGS results
  b. Heat cycling: relatively random
  c. Sonication: larger volume required, expensive machine, contamination risk
  d. Divalent cations: allow RNA to be self-cleaving (Mg^{2+}, Ca^{2+}, Zn^{2+}, …)
     - Mg^{2+} and heat (94°C) for 2-15 minutes depending on RNA quality
       - High quality RNA: long fragmentation time
       - Low quality RNA: short fragmentation time

RNA library prep

STEP 2. Fragmentation

Use shorter fragmentation times when RNA quality is compromised!

Fragmentation pattern of high quality RNA



Electropherogram from BioAnalyzer

Red	Ladder
Blue	150-300 bp, mRNA fragmented for 15 minutes at 94°C
Green	200-500 bp mRNA fragmented for 10 minutes at 94°C
Cyan	400-1,000 bp mRNA fragmented for 5 minutes at 94°C

RNA library prep

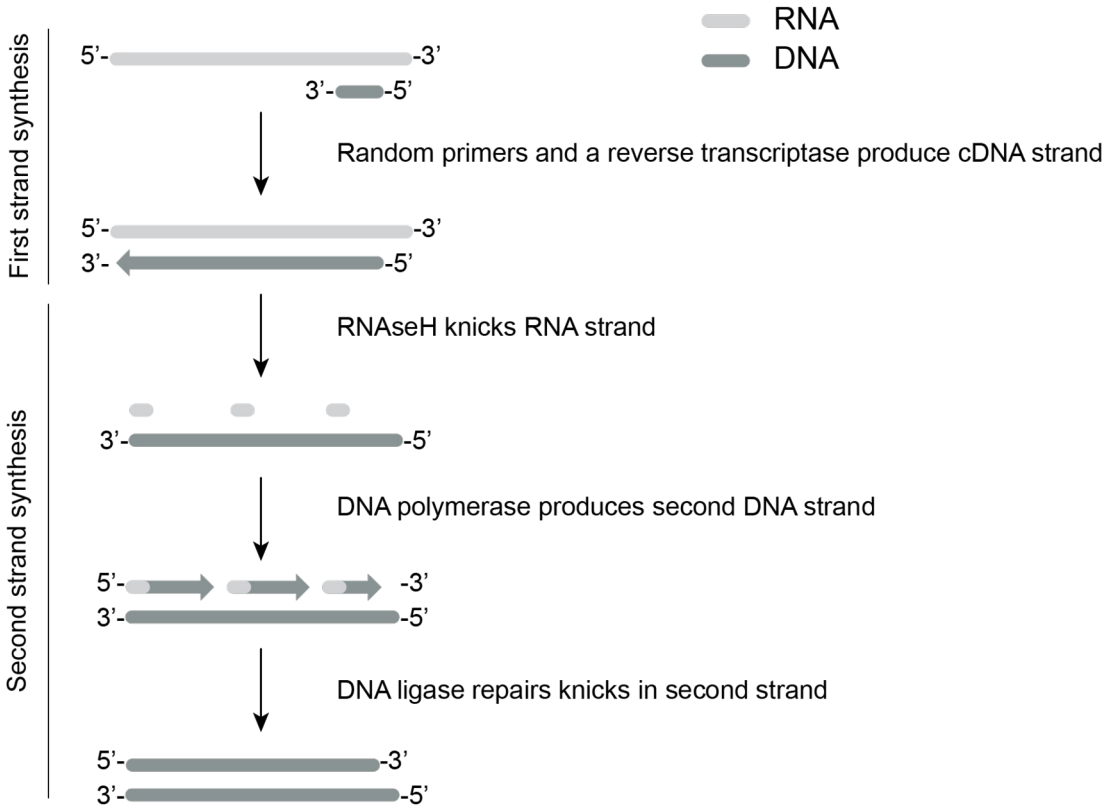
STEP 3. dsDNA synthesis

1. First strand synthesis
2. Second strand synthesis
3. Purify dsDNA by

SPRI bead clean-up

(1.8x)

*Depending on the kit, dsDNA is already end repaired

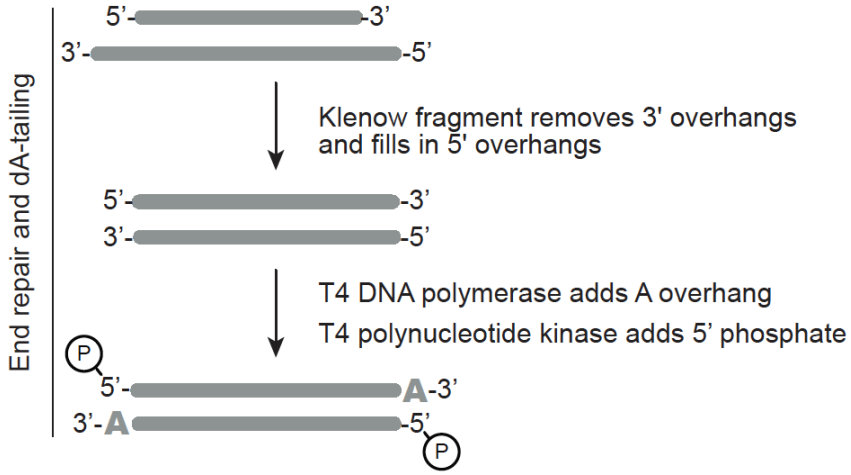


RNA library prep

STEP 4. End repair & dA-tailing

1. Remove overhangs to get blunt ends
2. Prep fragment to clone on adapters
   a. Add 5’ phosphate (T4 kinase)
   b. Add single 3’ A overhang to clone on adapters

(T4 DNA polymerase)



Note: After end-repair, move on to adapter ligation step ASAP

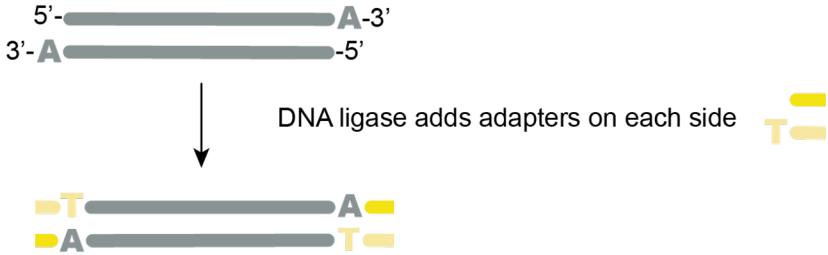
(5’ phosphate is fragile)

*Some kits obviate this step, the repair occurs during dsDNA synthesis

RNA library prep

STEP 5. Adapter ligation

1. Adapters are added to allow sequencing



1. Remove adapter dimers by SPRI bead clean-up (0.9x)

   Adapter dimers are our enemy #1 after this step (short fragments get preferentially sequenced)

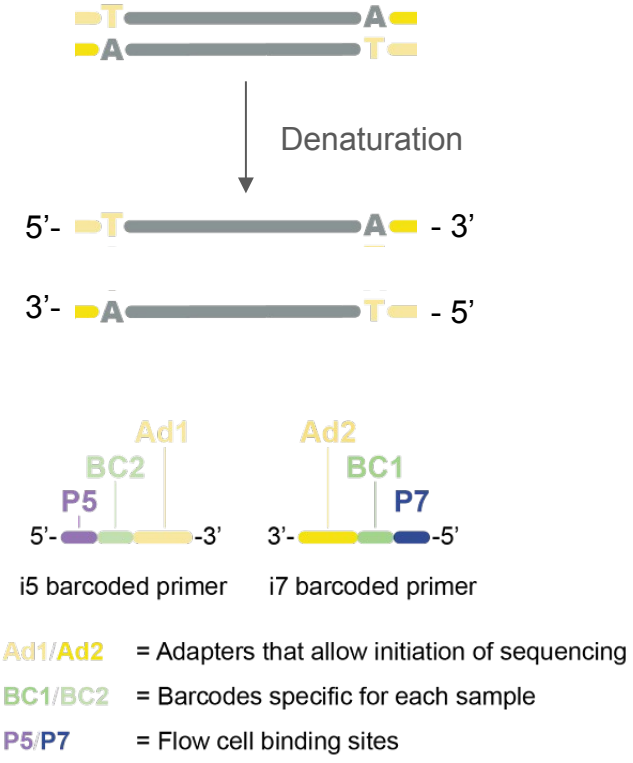


RNA library prep

STEP 6.

Barcoding PCR

1. i5 and i7 oligos are added by PCR, along with unique barcodes per sample
2. SPRI bead clean-up (0.8x)

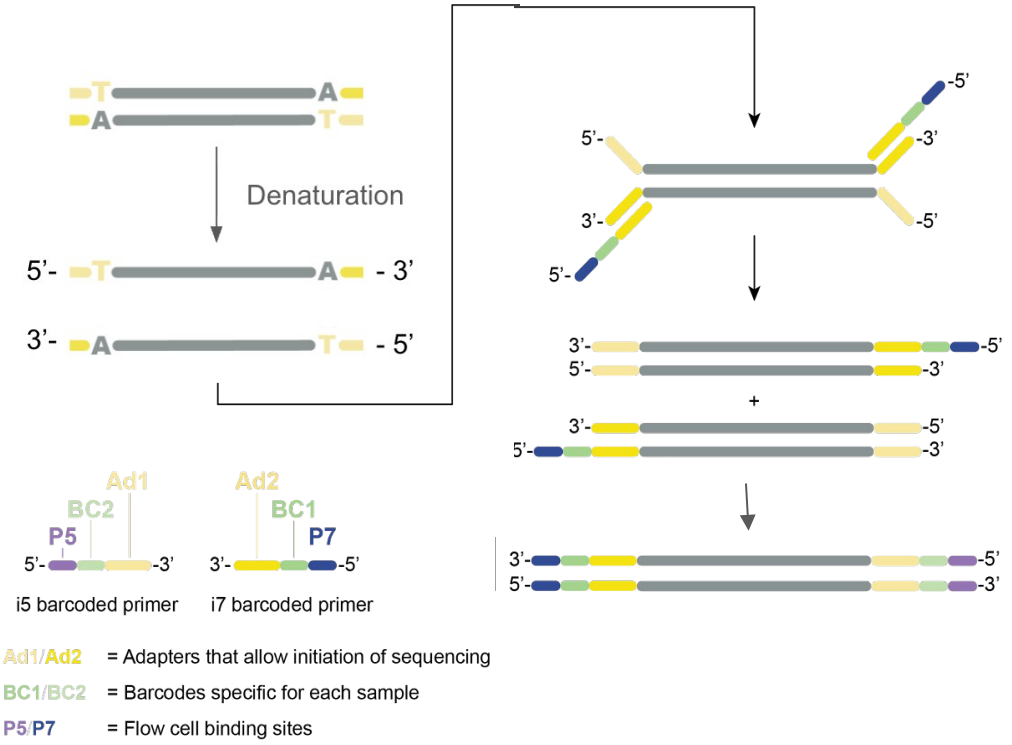


RNA library prep

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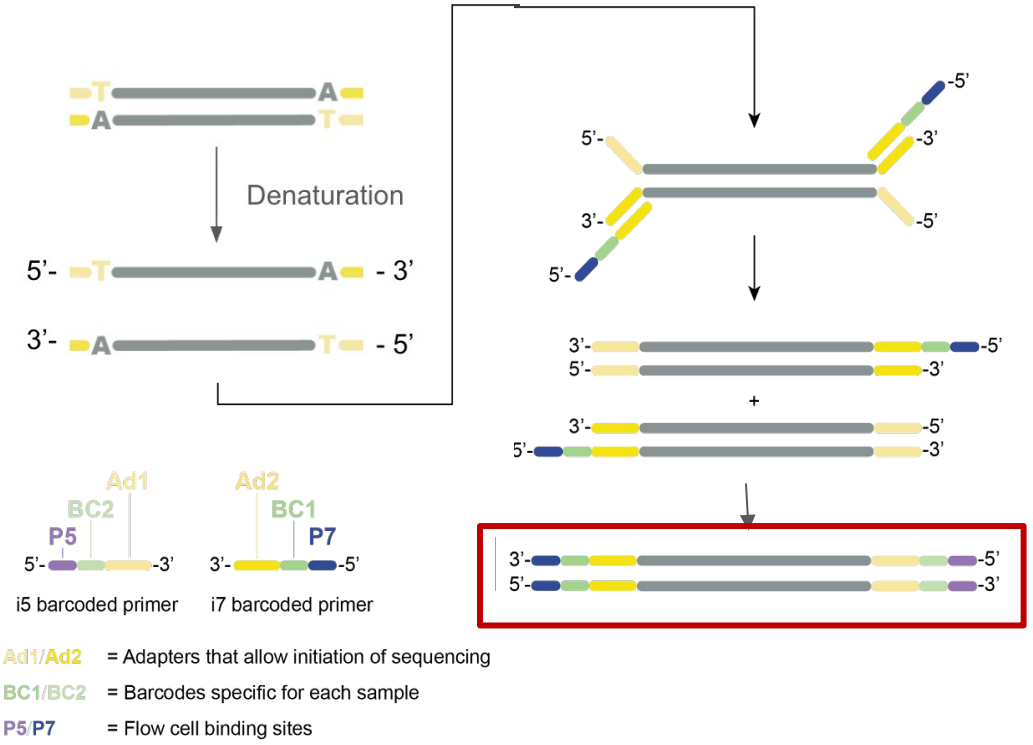


RNA library prep

STEP 6.

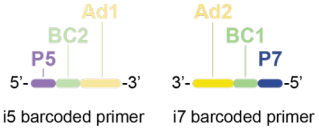
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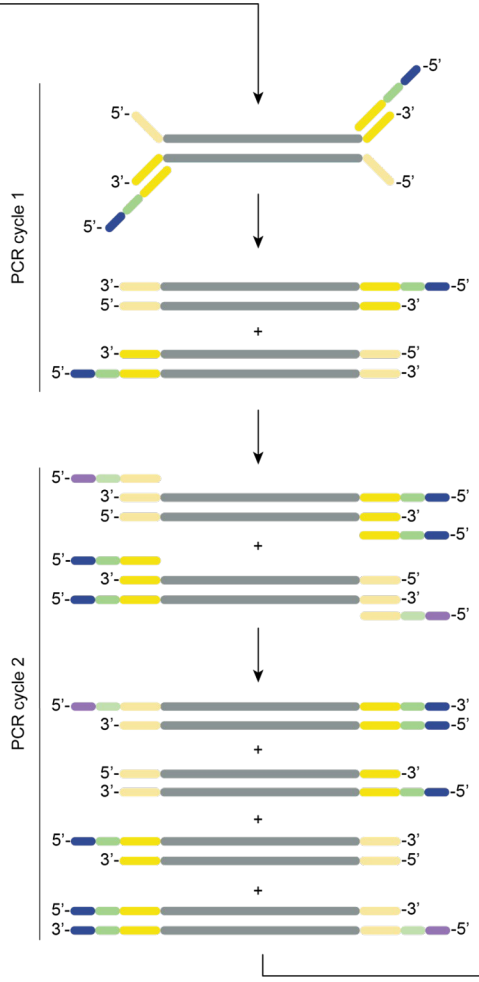




Ad1/Ad2 = Adapters that allow initiation of sequencing

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P5/P7 = Flow cell binding sites



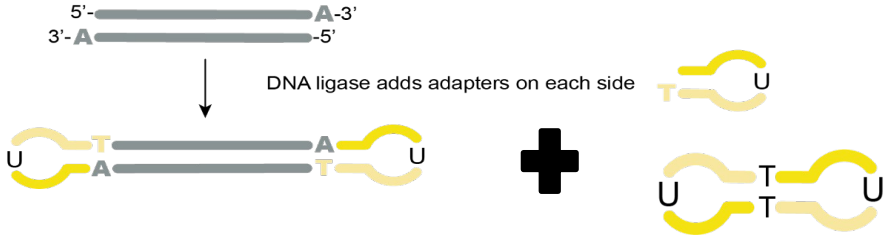
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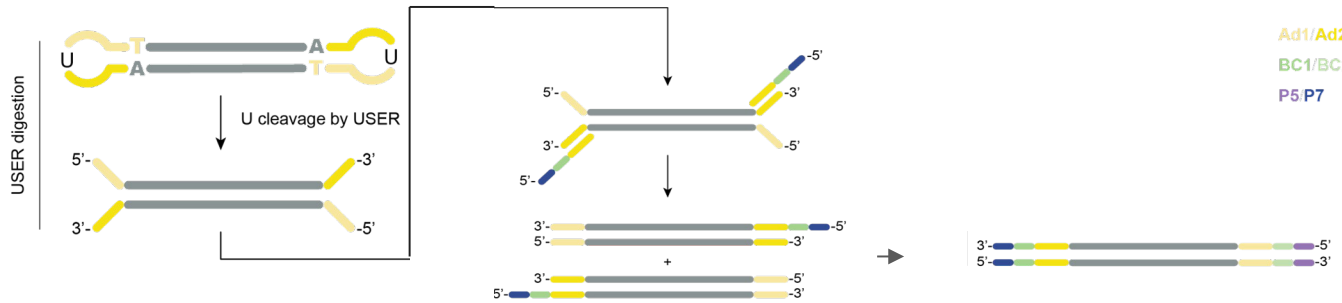
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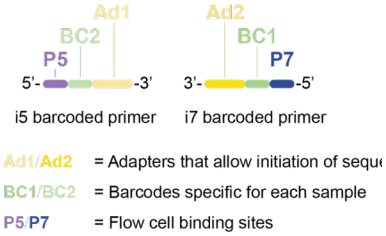
Note on alternative adapters, such as the NEB U shaped



Adapter dimers must be removed

Barcoding PCR starts with USER digestion



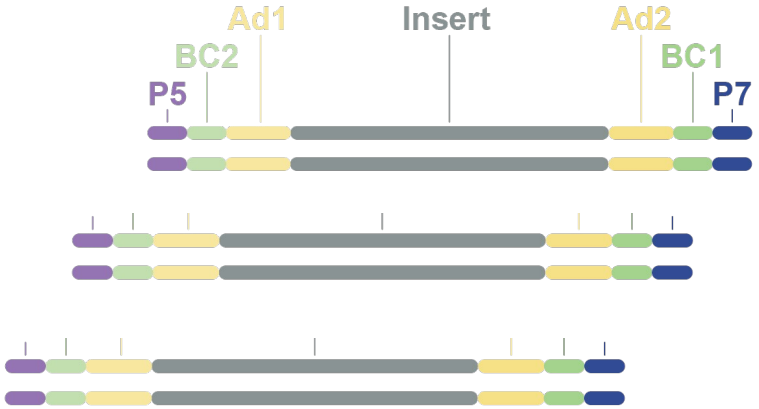


RNA to sequencing library



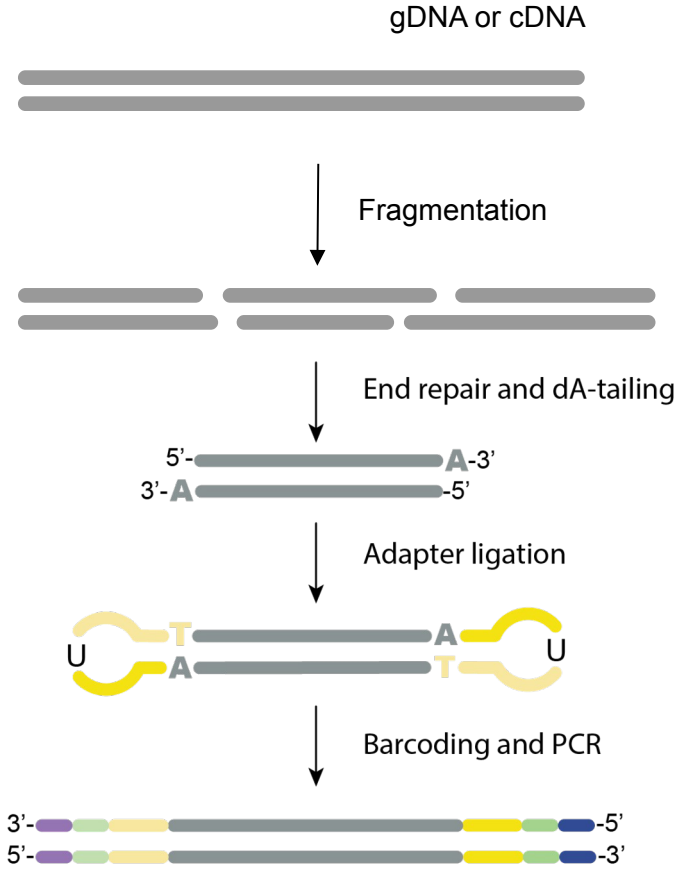
Extracted
RNA





Sequencing
library

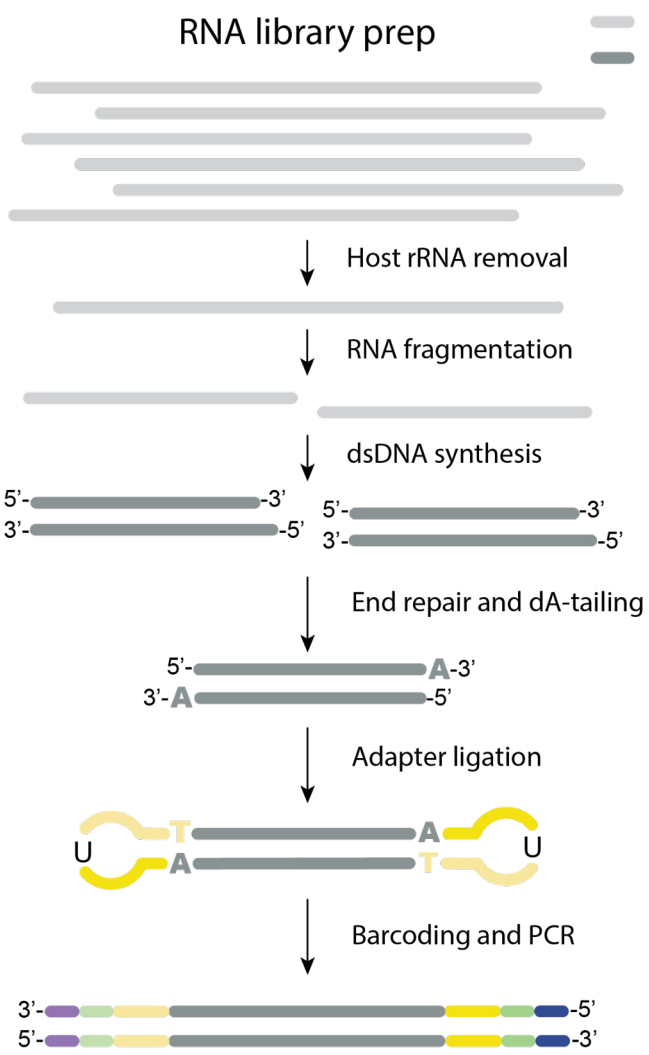
DNA library prep



DNA library kit options

	NEBNext Ultra II FS DNA	**Illumina DNA Prep** WGS, cDNA, amplicons	**Illumina Nextera XT** (small genomes, cDNA, amplicons)
	1 rxn, Fragm. enzyme + repair enzymes	1 rxn, Transposome frag. + ligation on bead	1 rxn, Transposome frag. + ligation in solution
	1 rxn, U shaped adapters		
	1 rxn	1 rxn	1 rxn

RNA library kit options



	NEBNext Ultra II RNA	Watchmaker RNA kit	Illumina rapid total RNA prep
	1 rxn, fragm. by Mg^{2+} & heat	1 rxn, fragm. by Mg^{2+} & heat	After rRNA depletion cDNA synthesis and tagmentation occur simultaneously ~3.5 hr
	1 rxn for 1st str. synt. 1 rxn for 2nd str. synt.	1 rxn for 1st str. synt. 1 rxn for 2nd str. synt.	
	SPRI bead cleanup		
	1 rxn	1 rxn, adapter ligation (miniadapters)	
		SPRI bead cleanup	
	1 rxn, U shaped adapters	1 rxn Barcoding PCR	
	SPRI bead cleanup	SPRI bead cleanup	
	1 rxn with an extra enzymatic step to digest the "U"		
	SPRI bead cleanup		

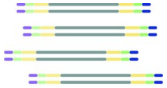
Watchmaker advantages

~4x input material

1 Rx less: 2ND SS & End repair are combined

1 less cleanup

The use of miniadapters leads to smaller AD and more efficient clean-ups



RNA library prep

Goal:

To make RNA and/or DNA Libraries that are compatible with Illumina sequencers

Workflow:

Very similar preparations between the 2 libraries but multiple steps required to make RNA into double stranded DNA.

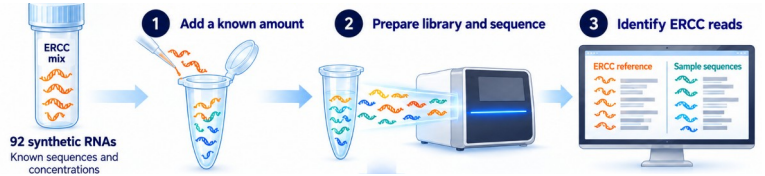
Considerations:

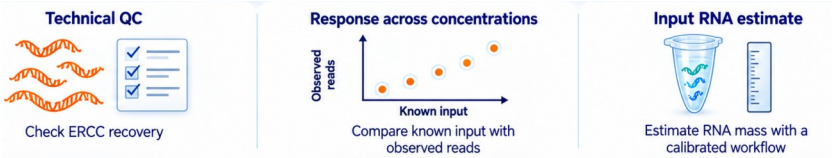
- How can we accurately assess composition of a sample of known input? ERCC spike ins
- How do we sequence many samples at once? Barcoding libraries and then pooling. Careful of mixing barcodes or samples pre-barcoding.
- Optimal insert length 400-500bp. Fragmentation of RNA/DNA is varied by time. How long for fragmentation to get the optimal insert size?
- To reduce PCR amplification bias, reduce PCR cycles to the minimal needed for sequencing (6-19 cycles)

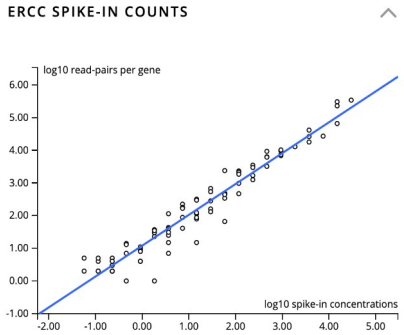
Synthetic standards for RNA library QC

ERCCs = External RNA Controls Consortium

- Pre-mixed, commercially available blend of 92 synthetic RNA transcripts
- Positive control for library preparation, spiked-in for all samples and water controls
- Can be used to estimate input RNA concentration







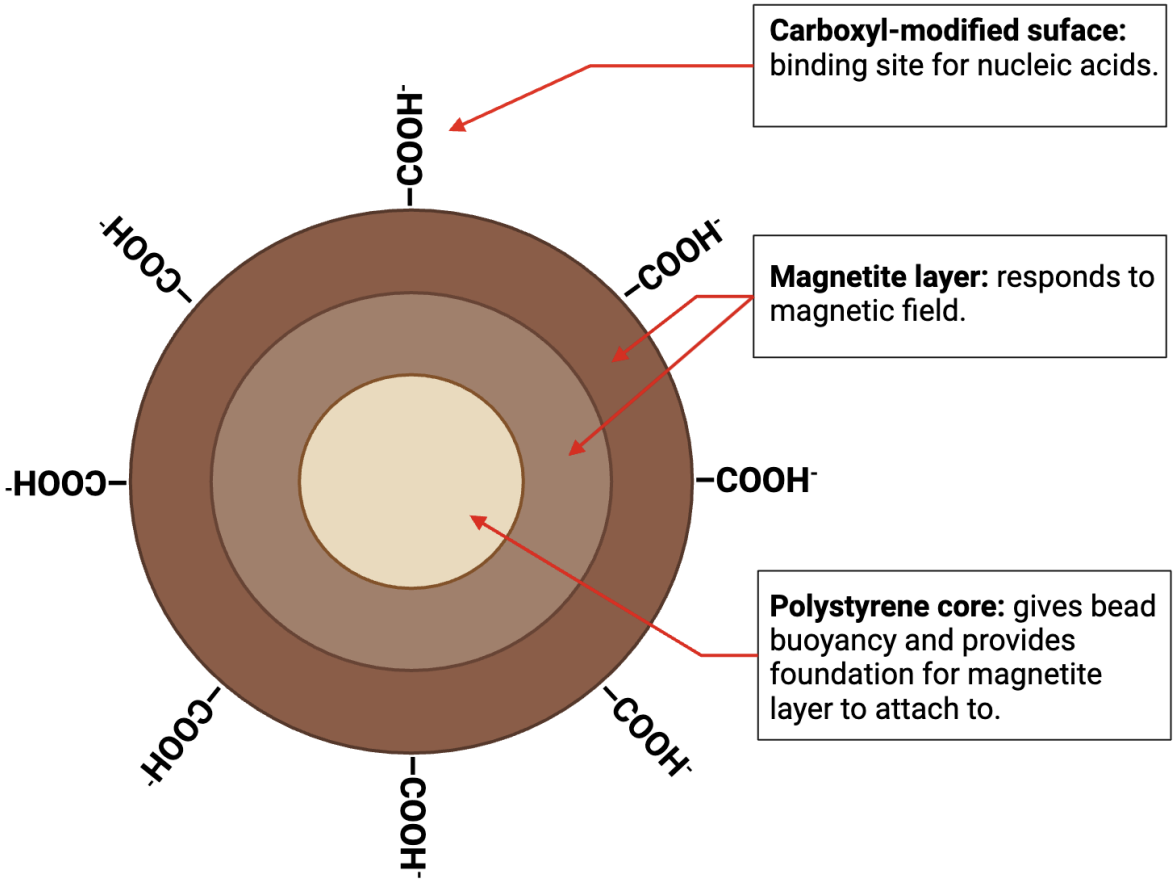
Plot generated from CZ ID

Introduction to SPRI Clean-ups

What are SPRI beads?

- Solid-phase Reversible Immobilization (SPRI) beads are paramagnetic beads used for **purification** and **size selection** of nucleic acids such as:
  - PCR amplicons
  - **NGS libraries**
  - **RNA/DNA**

What are SPRI beads?



How do SPRI beads work? *Purification*

- Under the right conditions, nucleic acids precipitate on the surface of the beads. The following are required:
  - **Crowding agent** to force nucleic acids out of solution.
  - **High salt concentration** to facilitate binding between nucleic acids and beads.
  - **Magnetic field induction** to control movement of nucleic acids.

Contents of SPRI buffer

Content	**Use**
Polyethylene Glycol (PEG) 8000	**Crowding agent** used to minimize space between nucleic acid and beads, favoring their proximity and the nucleic acid precipitating on the surface of the beads.

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Tris-HCl	**Regulates pH**.

Contents of SPRI buffer

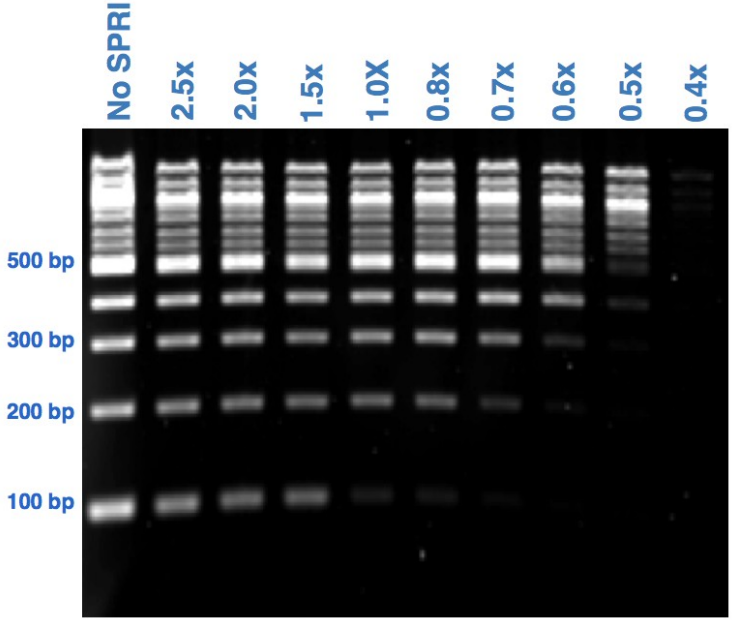
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Tris-HCl	**Regulates pH**.
EDTA	**Chelating agent** which sequesters ions such as Mg2+ and Ca2+ which could fragment/degrade DNA or activate any enzymes (e.g. nucleases) that are present in the sample. Aids in retaining the integrity of the sample.

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EDTA	**Chelating agent** which sequesters ions such as Mg2+ and Ca2+ which could fragment/degrade DNA or activate any enzymes (e.g. nucleases) that are present in the sample. Aids in retaining the integrity of the sample.
Tween 20	**Detergent** which decreases the adhesion of nucleic acids to the plastic walls of the tube by obstructing the ionic attraction.

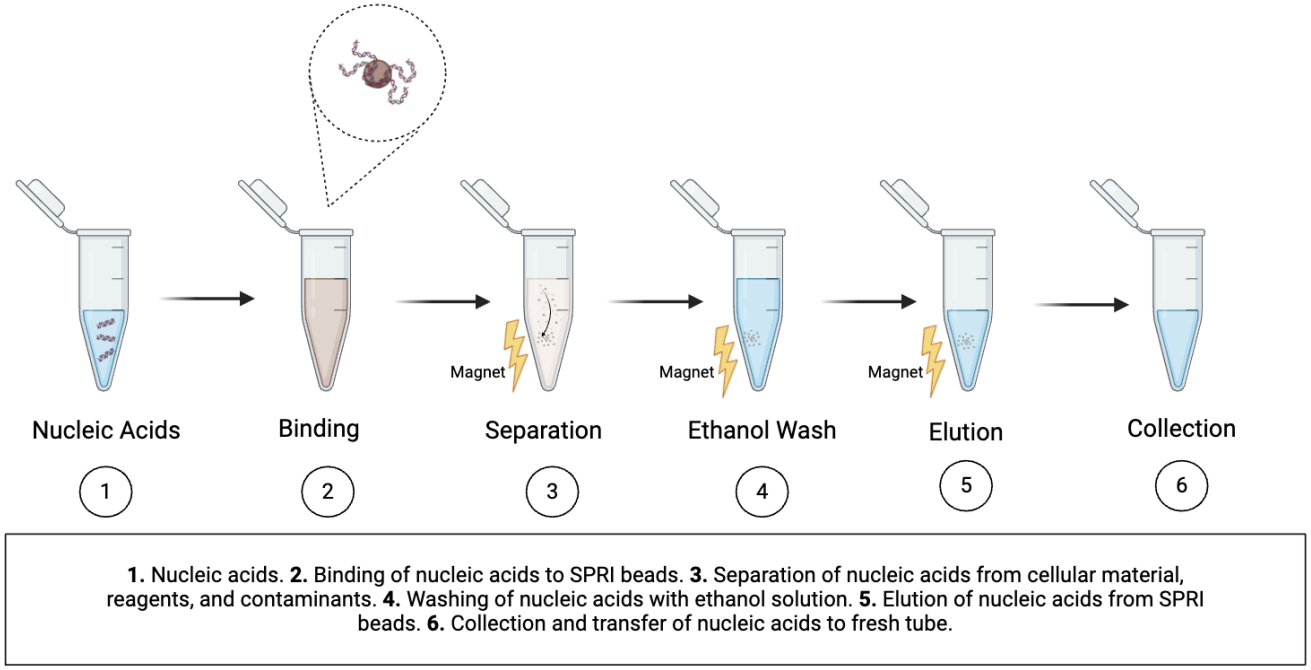
How do SPRI beads work? *Size selection*

- Size selection is dictated by the ratio between the volume of SPRI beads added to your sample and the volume of your sample.
    - **The higher the SPRI bead to sample volume ratio, the more smaller fragments will be retained.**
- For a desired SPRI ratio, to determine the volume of SPRI beads to add to your sample, multiply the SPRI ratio by the volume of your sample.



$$\text{SPRI ratio} = \frac{\text{Volume of beads}}{\text{Volume of reaction}}$$

Using SPRI beads in library preparation: *The process*

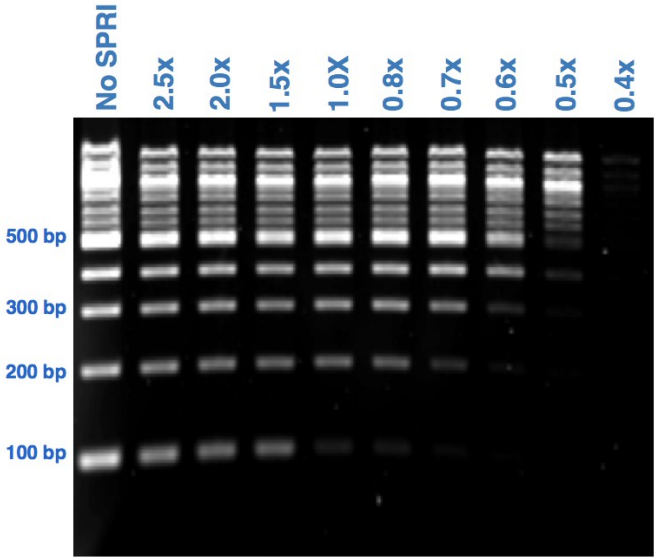


1. Nucleic acids. **2.** Binding of nucleic acids to SPRI beads. **3.** Separation of nucleic acids from cellular material, reagents, and contaminants. **4.** Washing of nucleic acids with ethanol solution. **5.** Elution of nucleic acids from SPRI beads. **6.** Collection and transfer of nucleic acids to fresh tube.

Note: Some commercial options are “SPRISelect Bead-Based Reagent” or “AMPure XP Beads”

Using SPRI beads in library preparation: *At which steps?*

Step	Ideal RNA/library size	To remove	SPRI ratio
After DNAse I treatment	-	Reagents	2x
After barcoding PCR	> 400 bp	Reagents and small fragments / dimers	0.6x
After pooling (if the pool needs to be concentrated	-	To concentrate the pool before loading	2x



$$\text{SPRI ratio} = \frac{\text{Volume of beads}}{\text{Volume of reaction}}$$

SPRI bead cleanup activity

Set up **2** reactions according to the table below at **0.8X** and **2.0X**. DNA input is 50bp DNA ladder diluted to 1:50 (save an aliquot to use as a control).

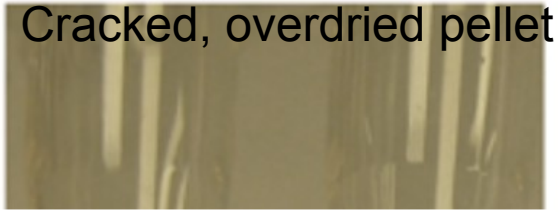
1. Add SPRI beads to sample and mix thoroughly by pipetting up and down ten times. Incubate for 3-5 mins at room temperature, then pulse spin the tubes for 1-2 seconds,
2. Place samples on magnetic rack and incubate for ~5 mins on the rack.
3. Remove supernatant.
4. Add 200µL of 85% EtOH to samples while on the magnetic rack. Do not disturb beads. Incubate at room temperature for 30s then remove the supernatant.
5. Repeat EtOH wash step for a total of 2 washes.
6. Air dry the beads for *up to* 3-5 mins while on the magnetic rack. *Watch beads carefully throughout this incubation to avoid overdrying; smaller bead pellets dry faster. Trust your eyes more than the timer.*
7. Remove tube from magnetic rack. Add 20ul nuclease free water to elute.
8. Pipette to mix well at least ten times. Incubate for a minimum of 2 mins at room temperature off the magnetic rack, then pulse spin the tubes for 1-2 seconds.
9. Place on magnetic rack until solution is clear ~5 mins.
10. Collect 18ul of the supernatant and transfer to a clean nuclease free PCR tube

Bead:Sample	0.8 : 1	2 : 1
50bp ladder 1:50 (µL)	20	20
SPRI beads (µL)	16	40

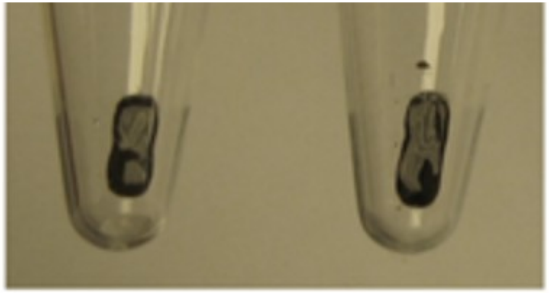
Do **not** overdry SPRI beads

Shiny, wet pellet

Cracked, overdried pellet

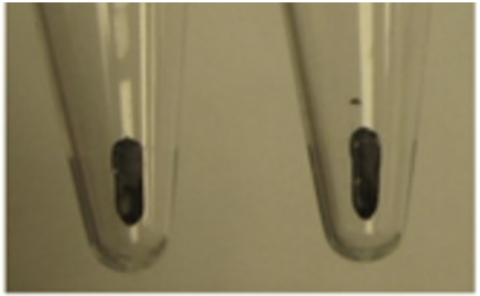


Risk of **EtOH carryover**





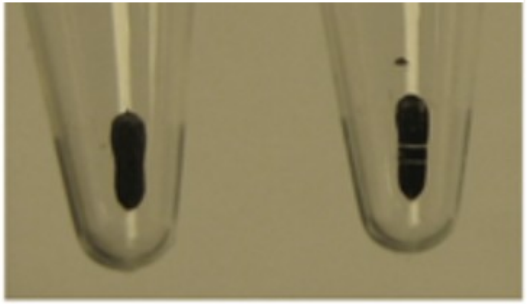
Ready for **elution**



Matte, dry pellet



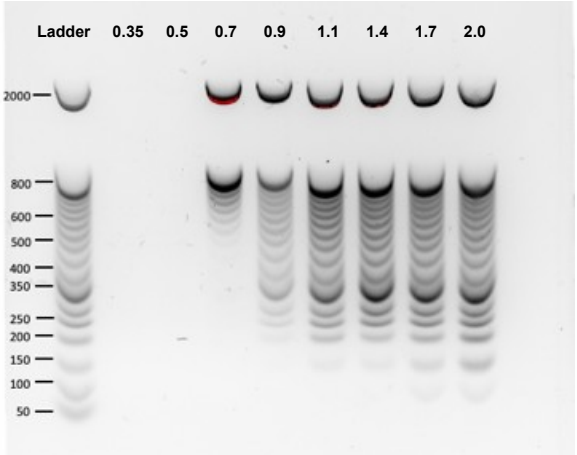
Risk of **low yield**



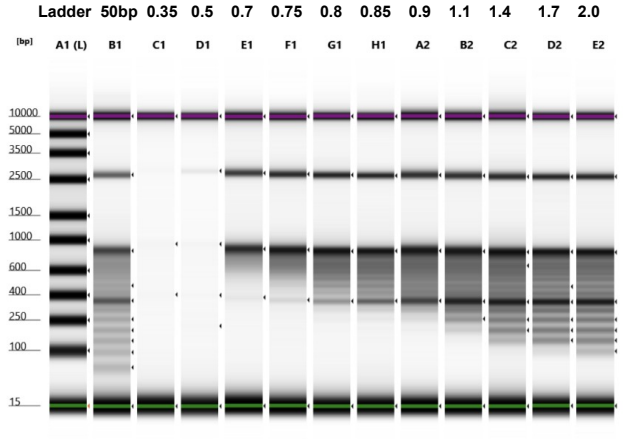
SPRI bead cleanup results

Quantify recovered sample by Qubit dsDNA HS

Visualize samples by agarose gel electrophoresis (1.8% agarose gel and 10000x SYBR GOLD DNA Stain or SYBR-Safe stain), capillary electrophoresis, or automated electrophoresis (TapeStation).



1.8% Agarose



TapeStation

Next: 05-I_Library prep QC for Illumina